

A PROJECT BRIEF · HACK THE HABITAT 2026

SafePassage

*A roadkill hotspot predictor and wildlife-crossing planner
for India's highways — built on real data, honest models,
and one simple question:*

where should the next crossing go?

COMPETITION

Hack the Habitat
Devpost · online · global

TEAM

Two students
ML & Data · Product & Web

BUILD WINDOW

Aug 24 – 30, 2026
Submit by Aug 31 · 10:15 AM IST

DOCUMENT

Version 1.0
Aug 23, 2026

WHAT FOLLOWS

Contents

1.	The Topic — roads are eating India's wildlife corridors	p. 2
2.	The Pitch — one line, four audiences	p. 3
3.	The Idea — from scattered sightings to ranked interventions	p. 3
4.	Scope — what we build, what we refuse to	p. 4
5.	Data & Attribution — sources, licenses, the honesty ladder	p. 4
6.	Architecture & the Contract — schema v1, API surface	p. 5
7.	The Stack — every choice, with its reason	p. 6
8.	The Division of Labour — two files, day by day	p. 6
9.	The Seven Days — an itinerary, not a wish	p. 7
10.	Judging Strategy — the rubric, the video, the checklist	p. 7
11.	Risks — named, owned, defused	p. 8
12.	Success, Metrics & the Road After	p. 9

SECTION ONE

The Topic — *roads are eating India's wildlife corridors*

India's national highway network — roughly 146,000 kilometres of it, and growing every budget cycle — pushes straight through tiger reserves, elephant corridors, and the Western Ghats, one of the world's eight biodiversity hotspots. Elephants are struck on railway tracks and highways in Kerala, Tamil Nadu, Assam, and Uttarakhand; leopards die on the Pune–Bengaluru stretch; uncounted reptiles, birds, and small mammals vanish under tyres without ever entering a record.

The deeper failure is not the deaths themselves — it is the silence around them. Most collisions are never reported. Official statistics lag reality by an order of magnitude. And when mitigation money

does flow — underpasses, fencing, signage, speed limits — it goes where construction is convenient or politics is loud, not where the animals actually die.

The raw material to fix this already exists. Citizen-science platforms — iNaturalist and its GBIF mirror — hold years of geotagged, time-stamped roadkill observations across India, uploaded by naturalists, students, and drivers. Road ecologists have proven the fix works: global studies report collision reductions of 80–90% when wildlife crossings are paired with fencing. What has never been built is the bridge between those two facts — a tool that turns scattered sightings into a ranked, fundable list of interventions. That bridge is SafePassage.

THE ANCHOR STORY — BANDIPUR, NH-766

The night-traffic ban on NH-766 through Bandipur Tiger Reserve — upheld by the Supreme Court in 2019 — ignited a running Kerala–Karnataka dispute: traders lose seven night-hours of highway, conservationists cannot prove which stretches actually kill. Both sides argue blind. SafePassage outputs the evidence layer that debate is missing, and gives our demo a story judges already know.

2

SECTION TWO

The Pitch — *one line, four audiences*

“*Show us your highway, and we'll tell you where animals will die next — and exactly what to build there to stop it.*”

SafePassage ingests every roadkill record citizen scientists have published for India, learns where and when wildlife-vehicle collisions concentrate, and hands forest departments, the NHAI, and local panchayats a ranked shopping list: *this* segment deserves the next crossing, *that* stretch needs painted signage and a seasonal speed limit, *this* cluster needs a fence first. Every recommendation is backed by visible data and an honest confidence score. Citizens and students feed the map by reporting sightings in seconds; researchers export the layers as GeoJSON for studies and litigation. It is decision support for the people who pour the concrete.

WHO ACTS ON IT	WHAT THEY GET
Forest departments & NHAI	A prioritized, cost-ranked list of segments for crossings, fencing, signage, and enforcement — evidence for budget meetings
Researchers & NGOs	Baseline mortality maps, seasonal curves, and exportable GeoJSON for studies, EIAs, and public-interest litigation
Citizens & students	A thirty-second sighting form and a live map of how their local hotspot is evolving — the data grows itself
Media & the public	Hard numbers for debates like Bandipur, instead of dueling anecdotes

3

SECTION THREE

The Idea — *four moves, one demo*

i. Ingest Real, citable data only — GBIF/iNaturalist roadkill records, the OpenStreetMap road graph, ESA WorldCover land cover, WDPA protected-area boundaries.	ii. Model Clean and join features; kernel-density hotspot mapping over observations; a gradient-boosted model scoring every road segment.	iii. Rank Segments scored 0–100 with seasonal collision curves, species mix, endangered-species flags, and an honest model confidence.	iv. Act Click any red segment and read a recommendation card — build a crossing, paint signage, enforce a limit — with its evidence attached.
---	---	--	---

THE DEMO MOMENT — DESIGNED, NOT HOPED FOR

A judge opens the link: India fades in, the Western Ghats glows. One click on the NH-766 corridor: the top five predicted segments appear, each with its season curve and a plain-language intervention. Total time to comprehension: **under ten seconds**. No login, no setup, no explanation needed — the product demonstrates itself.

4

SECTION FOUR

Scope — *what we build, what we refuse to*

IN SCOPE — THE MVP	DELIBERATELY CUT
— Interactive map — MapLibre GL, hotspot heatmap over the all-India road network — Segment intelligence — click any hotspot for risk score, species mix, season curve, confidence — Recommendation cards — crossing / fencing / signage / enforcement, with rationale — Seasonality calendar — when each segment is most dangerous, month by month — Species & taxa filters — elephants only, reptiles only, and so on — Report-a-sighting form — the crowdsourced growth loop, minimal and fast — Methodology page — live model metrics, evaluation stats, honest limitations — Attribution page — every dataset, license, and credit, in full	— User accounts & authentication — friction without value in seven days — Native mobile apps — the web app is responsive; that is enough — Multi-country expansion — one region done deeply beats three done shallowly — Real-time streaming feeds — daily batch is honest and sufficient — Training models from scratch — pretrained spatial statistics plus gradient boosting wins here — Any paid service or API — the entire project runs on free tiers and open data

DEFINITION OF DONE — PER MILESTONE

- **D2:** the deployed Vercel URL loads a map with fixture data — a judge could click it already
- **D4:** the pipeline notebook produces evaluation metrics and schema-valid GeoJSON v1 files
- **D5:** end-to-end click-through runs on real model output — integration declared closed
- **D6:** Lighthouse accessibility ≥ 90, mobile pass clean, loading and empty states handled
- **D7:** demo video (≤ 3 min) recorded and the Devpost entry drafted — nothing left to the buzzer

5

SECTION FIVE

Data & Attribution — *sources, licenses, honesty*

SOURCE	ROLE IN THE MODEL	LICENSE / CREDIT
GBIF · iNaturalist	Roadkill-flagged observations across India, ~10-year window, all taxa — the response variable	CCo / CC-BY per dataset — cited in-app and in the repo
OpenStreetMap	Road geometry, classes, junctions, proximity features via Overpass API	ODbL — attribution page + map credit
ESA WorldCover	10 m land cover — forest, plantation, water adjacency features	CC-BY 4.0

Data & Attribution — Honesty Ladder Result

Updates Section 5. The Day-1 09:00 honesty-ladder check (p.4) has been run against real iNaturalist data. This page records the result and the scope decision it triggers.

What we ran

GBIF's free-text search proved unreliable for this purpose — its *q* parameter does OR/token matching rather than phrase matching, so keyword queries like “road kill” and “hit by vehicle” matched millions of unrelated India records. We moved to iNaturalist's structured “Roadkill” observation field instead — a controlled annotation, not a keyword guess — queried directly via `field:Roadkill=Yes` for India (place_id 6903), and deduplicated by observation id across field-name variants.

The result

Source / query	Count
iNaturalist — structured field, deduplicated (primary signal)	92
iNaturalist — free-text “road kill” (secondary, cross-check)	95
GBIF — free-text “roadkill” (secondary, cross-check)	7,505*

**Single-word GBIF match, not phrase-restricted — kept only as an order-of-magnitude cross-check, not trusted as a clean count. It is broadly consistent with the iNaturalist structured count, which is the number we plan against.*

Ladder rung: RUNG 3 — pivot to evidence layer

92 nationwide records is under the 150-record threshold in our own honesty ladder (p.4). Per the ladder we pre-committed to, the predictive hotspot model is demoted from headline feature to a secondary, clearly-labeled, low-confidence layer. This is not a scope failure — it is the ladder working as designed, decided on Day 1 rather than discovered on Day 4.

What changes in the build

- **Headline feature** becomes the descriptive evidence layer: known, citable collision corridors (Bandipur/NH-766, Pune–Bengaluru, Assam elephant corridors) sourced from literature and news, not from the sparse dataset.
- The 92 iNaturalist points are shown as a secondary “citizen-science signal” layer, explicitly labeled sparse/low-confidence — no risk score is extrapolated into areas with zero coverage.
- Methodology page (p.4 scope, D6 milestone) states the record count and spatial coverage alongside any model metric, per our own honesty-ladder commitment.
- Next step before Day 1 model work begins: bucket the 92 observations by state to confirm whether regional pooling (Rung 2 language) is usable anywhere, or whether the evidence layer stands alone.

SECTION SIX

Architecture & the Contract — *where the two halves meet*

The entire collaboration hinges on one artefact: GeoJSON Schema v1, frozen on Day 1 and never broken afterwards. The frontend builds against fixture files conforming to it from hour one; the ML pipeline fills it with real output on Day 5. Integration becomes a formality, not a fire.



API SURFACE — SMALL ON PURPOSE

ENDPOINT	METHOD	PURPOSE
/api/hotspots	GET	Filtered hotspot features — bbox, species, min-risk; powers the map layer
/api/segments/:id	GET	Full segment dossier — season curve, species mix, recommendation, evidence
/api/stats/summary	GET	Headline metrics for the methodology page (top-5% capture, AUC, counts)
/api/sightings	POST	Report-a-sighting writes to Supabase; returns the new record

GEOJSON SCHEMA V1 — THE FROZEN CONTRACT (EXCERPT)

```
// safepassage.hotspots.v1 — one Feature per hotspot
{
  "type": "Feature",
  "geometry": { "type": "Point", "coordinates": [76.42, 11.67] },
  "properties": {
    "hotspot_id": "HS-000123",
    "risk_score": 87, // 0-100, ranked percentile
    "confidence": 0.82, // model-reported, shown in UI
    "species_mix": { "Aves": 12, "Mammalia": 4 },
    "endangered_flag": true,
    "season_curve": [3,2,2,4,6,9,12,10,7,5,3,2], // Jan..Dec
    "observation_count": 18,
    "nearest_highway": "NH-766",
    "intervention": "wildlife_crossing", // enum
    "model_version": "v0.3"
  }
}
```

A parallel `segments.v1` layer carries OSM way IDs and per-segment scores. Any change to either schema after Day 1 requires both leads to agree in writing in the commit message — the contract is the project's constitution.

SECTION SEVEN

The Stack — *every choice, with its reason*

LAYER	CHOICE	WHY THIS AND NOT SOMETHING ELSE
Frontend	React 18 + Vite, Tailwind CSS, MapLibre GL JS	Fastest scaffold-to-deploy; MapLibre is open-source (no Mapbox token, no billing surprise on demo day)
ML / Data	Python, geopandas, SciPy KDE, scikit-learn GradientBoosting	Battle-tested spatial stack; gradient boosting handles messy tabular features without deep-learning overhead
Backend	FastAPI + static GeoJSON tiles; Supabase free tier for sightings	Static-first serving survives traffic spikes and costs nothing; Supabase gives a real database in minutes
Hosting	Vercel (frontend + API), GitHub (public repo)	Live link from Day 1 — judges can click before the video even exists
Practices	Frozen schema v1 · fixture-first UI · daily 15-min sync	The coordination protocol that makes a two-person, seven-day build survivable

SECTION EIGHT

The Division of Labour — *two files, day by day*

FILE A — ML & DATA LEAD	FILE B — PRODUCT & FRONTEND LEAD
DAY 1-2 Run the 09:00 honesty-ladder check; download and clean GBIF/iNat records; join OSM roads and WorldCover features; co-freeze Schema v1 at the 10:00 ceremony; commit fixture files for the frontend. DAY 3-4 Build KDE hotspot maps and the gradient-boosted segment model; produce the headline stat — <i>"the top 5% of segments captures X% of recorded collisions"</i> — plus AUC and calibration for the methodology page. DAY 5 Swap real predictions behind Schema v1; stand up the thin FastAPI; hand over the summary metrics JSON. DAY 6-7 Write the methodology page content, the full attribution list, and the README; own half the Devpost writeup; protect the buffer.	DAY 1-2 Scaffold the React app; deploy the shell to Vercel <i>on day one</i> so the live link exists; build the full map UI against the committed fixture GeoJSON — never blocked, never waiting. DAY 3-4 Segment-click recommendation cards, seasonality calendar, species filters; design the ten-second demo path so it feels inevitable. DAY 5 Wire live model data behind the same schema; ship the report-a-sighting form on Supabase; declare integration closed. DAY 6-7 The polish pass — accessibility, mobile, loading and empty states; record the demo video; own half the Devpost writeup.

9

SECTION NINE

The Seven Days — *an itinerary, not a wish*

Mon · Aug 24	Day 1 — Evidence & Contract. 09:00 honesty-ladder data check; schema freeze at 10:00; fixtures committed by evening; app shell deployed to Vercel before dinner.	BOTH
Tue · Aug 25	Day 2 — Parallel Tracks. Feature engineering deepens; map UI runs fully on fixtures; the live link behaves like a product already.	A · B
Wed · Aug 26	Day 3 — The Model. KDE hotspots take shape; segment risk model training begins; frontend builds cards and the seasonality calendar.	A · B
Thu · Aug 27	Day 4 — Numbers That Convince. Model frozen with evaluation stats; headline metric computed; demo path rehearsed on fixture data.	A · B
Fri · Aug 28	Day 5 — Integration, the Non-Event. Real predictions behind the same schema by 11:00; FastAPI live; sightings form shipped; end-to-end click-through green.	BOTH
Sat · Aug 29	Day 6 — Polish Is the Product. Accessibility ≥ 90, mobile pass, loading/empty states; methodology and attribution pages written; screenshots captured.	A · B
Sun · Aug 30	Day 7 — Ship Ahead of the Gun. Demo video (≤ 3 min) recorded and edited; Devpost entry drafted, reviewed by both leads, ready to submit.	B · A
Mon · Aug 31	Submission morning. Final review, submit before 10:15 AM IST — with hours to spare, because submission logistics eat teams alive at the buzzer.	BOTH

10

SECTION TEN

Judging Strategy — *answer the rubric on purpose*

CRITERION · WEIGHT	OUR DELIBERATE ANSWER
Environmental Impact · 30%	A ranked intervention list converts death data into <i>prevented</i> deaths — anchored on the live Bandipur controversy, with a named stakeholder path (forest dept / NHAI)
Use of Technology · 25%	Real spatial statistics and gradient boosting on messy citizen-science data — not another ChatGPT wrapper
Design & Usability · 20%	The ten-second demo path; a judge clicks one hotspot and receives an action; accessibility treated as a requirement, not a wish
Execution · 15%	Live link from Day 1; integration scheduled as a non-event; a buffer day before the deadline
Theme Alignment · 10%	The output <i>is</i> protection — every recommendation maps to a physical, fundable, life-saving measure

THE DEMO VIDEO — THREE MINUTES, FIVE BEATS

TIME	BEAT
0:00–0:20	The hook. Bandipur at night — the ban, the dispute, the blindfold. "Both sides are arguing without a map. We built the map."
0:20–0:50	The problem & the data. Highways vs. corridors; the sightings that already exist and nobody uses. Show the raw records — real, citable, licensed.
0:50–1:50	The live demo. The ten-second path: map → hotspot → recommendation card → season curve → report-a-sighting. No cuts, no mocks.
1:50–2:30	The honest model. Methodology page on screen: metrics, confidence scores, limitations stated plainly. The honesty ladder, explained.
2:30–3:00	The ask. What a forest department does Monday morning with this list; the roadmap; the team. End on the one-liner.

THE DEVPOST CHECKLIST — NOTHING MISSING, NOTHING LATE

- ☐ Working demo — live Vercel link *plus* a recorded video (we submit both)
- ☐ Public source repository — clean commit history, README with setup instructions
- ☐ Written project description — problem, implementation, technologies (drafted Day 7, not Day 7:30)
- ☐ Clear attribution — every dataset, library, and API credited in-app and in-repo
- ☐ Submission completed **before Aug 31, 2026 · 10:15 AM IST** — target: Aug 30 evening

SECTION ELEVEN

Risks — *named, owned, defused*

RISK	OWNER	DEFUSAL
India data too sparse	ML	The honesty ladder — pre-decided fallbacks, executed at 09:00 Day 1, never a mid-build panic
GBIF download quirks	ML	Day-1 spike with CSV bulk-download fallback; no API dependency for the core dataset
MapLibre styling time-sink	Web	Timebox 4 hours; fallback to Leaflet + heatmap plugin, which we already know cold
Integration drift	Both	Frozen schema v1 + committed fixtures + a five-line validation script run on every data commit
Scope creep	Both	The cut list on page 4 is a treaty, not a suggestion; new ideas go to the roadmap, not the sprint
Supabase friction	Web	Fallback: embedded Google Form for sightings — ugly but shippable in twenty minutes

Risks — Update to “India Data Too Sparse”

Updates Section 11 (p.8). The risk originally listed as a possibility is now a confirmed, quantified condition, with its defusal already executed rather than pending.

Risk	Owner	Status	Defusal
India data too sparse	ML	CONFIRMED	Honesty ladder executed Day 1: 92 structured records → Rung 3. Evidence layer is now the headline feature; predictive model demoted to a labeled secondary layer.

A related finding worth naming as its own risk

Free-text search on GBIF/iNaturalist is unreliable for this data. During the check, plain keyword queries returned OR-matched, wildly inflated counts (millions of records for “road kill” on GBIF) rather than true phrase matches. Any future ingestion step that relies on free-text search instead of structured fields will silently reproduce this noise. Owner: ML. Defusal: ingestion pipeline uses iNaturalist’s structured observation-field query exclusively; free-text results are retained only as an order-of-magnitude cross-check, never summed into the primary count.

Why this doesn’t sink the entry

- The rubric rewards an honest, defensible confidence story as much as a clean metric (Environmental Impact 30%, Use of Technology 25%) — a visible “low data” zone reads as more credible to a judge than a suspiciously confident map.
- The Bandipur/NH-766 anchor story and the evidence-layer approach were already the emotional core of the pitch (p.2–3); this shifts engineering emphasis, not the narrative.
- The decision was made against pre-written thresholds on Day 1, exactly as planned — not improvised under deadline pressure.

12

SECTION TWELVE

Success, Metrics & the Road After

- **Success means:** a judge understands the problem, the evidence, and the action in one visit — and believes a forest department could too.
- **Metrics we hold ourselves to:** top-5% of segments capturing a large share of recorded collisions · Lighthouse ≥ 90 · demo path under ten seconds · zero paid services · every source credited.
- **The road after the hackathon:** pilot outreach to one forest division with the NH-766 segment list · a WhatsApp reporting bot (where Indian citizens actually are) · Hindi and Kannada localization · a crossing cost-benefit calculator · a railway-sector module for elephant corridors.

Prepared by the SafePassage team — two students, one contract, seven days, and a map that tells the truth. · v1.0 ·

Aug 23, 2026 · docs: E:\Projects\hack-the-habitat